How Stable Are Regression Conclusions Across Subpopulations? A Finite-Sample Empirical Study

Anonymous

Abstract

Pooled linear regression is often used to summarize associations across heterogeneous subpopulations, although a pooled coefficient need not describe every subgroup equally well. This paper studies the finite-sample stability of pooled regression conclusions by comparing pooled and subgroup-specific ordinary least squares fits. We consider coefficient signs, standardized departures from the pooled estimate, covariate-context adjustment, and loss of predictive accuracy within subgroups. A unified simulation varies group count, total sample size, group-size imbalance, covariate shift, and coefficient heterogeneity across four data-generating scenarios. Even under common slopes, separate subgroup estimates diverge when groups are small or unevenly sized. Covariate shift produces large raw distributional differences but little additional predictive loss, whereas coefficient heterogeneity produces persistent coefficient divergence and poorer subgroup prediction. When both mechanisms are present, raw drift and prediction loss are largest. An age-stratified empirical illustration shows that an aggregate regression can conceal subgroup-specific patterns. The paper presents a diagnostic framework for evaluating whether a pooled regression is an adequate summary across observed subpopulations; it does not propose a new estimator or a test of causal effect modification.

Keywords: regression diagnostics; subpopulation heterogeneity; coefficient stability; covariate shift; Monte Carlo simulation; model transportability

1 Introduction

Researchers often fit one linear regression to data pooled across observed groups, such as age categories, regions, treatment arms, or demographic groups. Pooling uses all available observations and can improve precision. It also imposes a substantive restriction: the resulting coefficient vector must serve as an adequate summary for all groups. If the underlying relationships differ, a pooled coefficient may describe no group particularly well. If groups are small or have different covariate distributions, subgroup estimates may also disagree even when their population slopes are the same.

This distinction matters because coefficient disagreement has several possible explanations. Under correct linear specification with common slopes, changing the covariate distribution affects the precision of subgroup estimates through the information matrices $X_g^\top X_g$, but it does not by itself change the conditional expectation of the subgroup OLS slope. Under misspecification, however, an OLS coefficient is a best linear approximation that depends on the regressor distribution (Berk et al., 2019). In that setting, distributional differences can change the population linear approximation even without a change in the underlying conditional mean function. A pooled analysis therefore requires attention both to sampling variation and to the relation between the fitted linear model and the covariate distribution.

Existing approaches address related but narrower questions. Structural-break procedures study coefficient stability in ordered data (Garbade, 1977; Watson and Engle, 1985). Heterogeneous-treatment-effect methods examine whether a particular treatment effect varies across groups (Wager

and Athey, 2018). Tsoi and Sun (2026) compare interaction and stratified regressions for subgroup treatment-effect analyses. Stable regression methods seek coefficients that are less sensitive to random splits (Bertsimas and Paskov, 2020), while latent-group models estimate unobserved group structure (Liu et al., 2020). The present setting is different: group membership is observed, the model is cross-sectional, and the question is whether an already fitted pooled regression remains an adequate descriptive and predictive summary within those groups.

This paper develops a finite-sample diagnostic study of that question. We compare pooled and subgroup-specific ordinary least squares fits using five quantities: sign stability, standardized coefficient drift, context-normalized drift, the pooled-versus-subgroup prediction gap, and covariate shift. A controlled simulation separates finite-sample variation, covariate shift, coefficient heterogeneity, and their combination. An age-stratified empirical illustration then shows how the diagnostics behave in observed data. The contribution is diagnostic rather than estimative. It does not introduce a new regression estimator or establish a causal test for effect modification.

Section 2 defines the diagnostics and model comparisons. Section 3 describes the simulation and its results. Section 4 presents the empirical illustration. Section 5 discusses the mathematical interpretation and limitations.

2 Methods

Let Y denote the response, let $X \in \mathbb{R}^p$ denote the predictors, and let $g \in \{1, \ldots, G\}$ index observed subgroups. The subgroup sample size is n_g, the subgroup coefficient for predictor j is $\hat{\beta}_{j,g}$, and the pooled coefficient is $\hat{\beta}_{j,\text{pool}}$. We treat the pooled regression as the estimand to be assessed, not as a benchmark assumed to be correct. The analysis records five complementary diagnostics of subgroup disagreement.

2.1 Core Stability Metrics

Table 1 summarizes the five primary metrics used in the paper.

Table 1: Core stability metrics used in the study

Metric	**Formula**	**Interpretation**
Sign stability rate (predictor j)	$I_{j,m} = \mathbf{1}(\operatorname{sgn}\hat{\beta}_{j,a_m} = \operatorname{sgn}\hat{\beta}_{j,b_m})$, $\text{SSR}_j = \frac{1}{M}\sum_{m=1}^{M} I_{j,m}$.	Proportion of comparisons with the same effect direction.
Standardized coefficient drift (predictor j)	$D_j = \max_g \left\lvert \hat{\beta}_{j,g} - \hat{\beta}_{j,\text{pooled}} \right\rvert$, $D_j^\star = D_j / \widehat{\text{SE}}\left(\hat{\beta}_{j,\text{pooled}}\right)$.	Largest subgroup departure from pooled effect, scaled for cross-predictor comparability.
Context-normalized drift (predictor j)	$M_j = \frac{1}{\binom{G}{2}} \sum_{a<b} \frac{\lvert \bar{x}_{j,a} - \bar{x}_{j,b} \rvert}{s_{j,\text{pooled}}}$, $\widetilde{D}_j = \frac{D_j^\star}{1+M_j}$.	Coefficient drift scaled by subgroup mean separation, so large drift is interpreted relative to observed covariate spread.
Pooled-versus-subgroup prediction gap (g)	$\text{TPG}_g = \text{RMSE}(\hat{f}_{\text{pool}}; g) - \text{RMSE}(\hat{f}_g; g)$.	Loss from using the pooled fit rather than the group-specific fit on observations in group g.
Covariate shift summary between groups a and b	$\text{CSS}_{a,b} = \frac{1}{p}\sum_{j=1}^{p} \frac{\lvert \bar{x}_{j,a} - \bar{x}_{j,b} \rvert}{s_{j,\text{pooled}}}$.	Average standardized difference in predictor distributions across groups.

For SSR_j, the pairs (a_m, b_m) are all pairwise comparisons of subgroup-specific coefficient signs. The prediction gap is evaluated within each target group and then averaged across groups; it is not a pairwise source-to-target prediction experiment.

Higher values of $D_j^\star$, $\widetilde{D}_j$, $\mathrm{TPG}_{a\to b}$, and $\mathrm{CSS}_{a,b}$ indicate greater instability or mismatch, while higher SSR_j indicates greater directional stability.

2.2 Regression Models Compared

For each dataset or simulated replicate, we fit four linear regression specifications. The pooled model is

$$Y = \beta_0 + X^\top \beta + \varepsilon,$$

which imposes a single coefficient vector across all subgroups. The second adds a group indicator,

$$Y = \beta_0 + X^\top \beta + \gamma G + \varepsilon,$$

which allows subgroup-specific intercept shifts but keeps slopes common. The third adds interactions,

$$Y = \beta_0 + X^\top \beta + \gamma G + X^\top \delta G + \varepsilon,$$

which allows the slope coefficients to vary by subgroup. The fourth fits separate subgroup-specific OLS models, so that each group has its own intercept and slope vector.

The pooled and separate models provide the quantities used in the diagnostics. The group-indicator and interaction models provide conventional reference specifications for intercept and slope differences. The comparison is descriptive; no single model is treated as the universally correct alternative.

2.3 Simulation Design

The simulation starts with a common-slope baseline and then varies the source of subgroup disagreement. It changes total sample size, the number of groups, group-size imbalance, and whether predictor means or regression coefficients differ across groups. The predictor correlation is fixed at $\rho = 0.2$, the outcome noise standard deviation is one, and there are two predictors. The four scenarios are: common predictor distributions and slopes; shifted predictor means with common slopes; common predictor distributions with shifted slopes; and simultaneous shifts in means and slopes.

For every replicate, we generate data, fit the four regressions, and retain the pooled and subgroup-specific coefficients and prediction errors. Each reported setting-level result is the average of replicate-level diagnostics. The simulation is designed to characterize finite-sample behavior under these specified mechanisms, not to estimate a universal threshold for deciding whether a pooled model should be used.

2.4 Interaction-Test Benchmark

We compare the diagnostic framework with the conventional test for slope interactions. The benchmark compares the group-intercept model to the model with group-specific slopes using the nested-model F-test at level $\alpha = 0.05$. It is run on a reduced grid with $G \in \{3, 15, 40\}$, $n \in \{300, 3000, 30000\}$, and balanced or random group-size allocation. The four data-generating scenarios are retained, and each scenario-setting cell is replicated 200 times. The benchmark reports the proportion of replicates rejecting common slopes. It is intended as a reference comparison rather

than a replacement for the descriptive diagnostics, which quantify coefficient movement, covariate separation, and pooled-versus-subgroup prediction loss.

2.5 Real-Data Workflow

The empirical analysis uses the same workflow. After defining the response, predictors, and subgroup variable, we fit the pooled, group-indicator, interaction, and separate subgroup regressions. We then compare coefficient diagnostics and the pooled-versus-subgroup prediction gap within each group.

To assess predictive performance beyond the estimation sample, we also use repeated stratified cross-validation. Within each repetition, observations are assigned to folds separately within each subgroup. A pooled model is fitted on the combined training observations, while a separate model is fitted for each subgroup using its training observations. Both models are evaluated on the held-out observations from that subgroup. The reported RMSE difference is $\mathrm{RMSE}_{\mathrm{pooled}} - \mathrm{RMSE}_{\mathrm{group}}$, so positive values favor the group-specific model. The empirical analysis remains illustrative. It does not identify causal effect modification or establish that the observed subgroup patterns generalize beyond the selected specification.

2.6 How the Metrics Are Used

Sign stability asks whether coefficient directions agree. Standardized drift measures the largest subgroup departure from the pooled coefficient on a common scale. Context-normalized drift places that departure in relation to observed subgroup mean separation. The prediction gap measures the within-group cost of using the pooled model rather than the group-specific model. Covariate shift summarizes differences in predictor means. These quantities should be read jointly. None is a formal test of a common slope, and none on its own identifies the source of subgroup disagreement.

3 Simulation Study

The simulation examines how the diagnostics behave when subgroup differences arise from known mechanisms. It holds the basic linear model fixed and varies group count, total sample size, group-size allocation, predictor means, and slopes. The design is intentionally low-dimensional so that changes in the diagnostics can be traced to the data-generating mechanism.

3.1 Simulation Design

3.1.1 Data-generating process

For each simulated dataset, we generate $p = 2$ predictors (x_1, x_2) from a bivariate normal distribution with means (μ_1, μ_2) and covariance matrix $\Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$. The response y follows a linear model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma_y^2).$$

The baseline coefficients are $\beta = (1,\, 0.8,\, -0.5)$, the predictor correlation is set to $\rho = 0.2$ (near-independence), and the noise standard deviation is $\sigma_y = 1$.

3.1.2 Four heterogeneity scenarios

The four scenarios separate distinct sources of subgroup disagreement.

Scenario 1 (no heterogeneity). All G groups share the same predictor distribution and coefficients. Any observed coefficient differences arise purely from finite-sample variation and mechanical differences in the design matrix ($X_g^\top X_g$) across groups.

Scenario 2 (covariate shift only). The group-specific predictor means $\mu^{(g)}$ are shifted by up to 0.5 standard deviations, but the coefficients $\beta^{(g)}$ remain identical across groups. This isolates the effect of covariate distribution differences on stability metrics.

Scenario 3 (coefficient heterogeneity only). The group-specific coefficients are shifted linearly across groups while predictor distributions are held constant. For x_1, the shift ranges from -0.4 to 0.4 across groups; for x_2, it ranges from 0.2 to -0.2. Given the baseline slopes, these shifts change coefficient magnitudes but retain their population signs. Sign reversals in this design can therefore arise from finite-sample estimation rather than being imposed by construction.

Scenario 4 (combined heterogeneity). Both predictor means and coefficients vary across groups, combining the shifts from Scenarios 2 and 3. This represents the most realistic case where groups differ in both their covariate distributions and their regression relationships.

3.1.3 Imbalance patterns

Group sizes are allocated according to four imbalance patterns.

Balanced: all groups receive equal sample sizes.

Mild: group sizes follow a shallow linear weight sequence from 1.2 to 0.8.

Strong: group sizes follow a steep linear weight sequence from G down to 1, producing one large group and one very small group.

Dirichlet-random: group proportions are drawn from a flat Dirichlet$(1, \ldots, 1)$ distribution, equivalent to normalizing G independent Exponential(1) draws by their sum. This pattern samples a wide range of allocations, from near balance to strong concentration, without imposing a fixed ordering of group sizes.

A minimum floor of 5 observations per group is enforced to avoid degenerate subgroup fits; the floor is lowered dynamically to $\lfloor n/G \rfloor$ when $G \times 5$ exceeds the total sample size.

3.1.4 Settings grid

The simulation grid varies four factors:

- Number of groups: $G \in \{2, 3, 5, 15, 40\}$.
- Total sample size: $n \in \{100, 300, 1200, 3000, 6000, 30000, 200000\}$.
- Imbalance pattern: $\{\text{balanced}, \text{mild}, \text{strong}, \text{Dirichlet-random}\}$.
- Scenario type: $\{\text{S1}, \text{S2}, \text{S3}, \text{S4}\}$.

Each scenario contains $5 \times 7 \times 4 = 140$ settings, for 560 settings in total. Settings with fewer than three observations in at least one subgroup are excluded from aggregate summaries. This leaves 136 directly comparable settings per scenario.

Each setting is replicated 200 times. Within each replicate, all four regression specifications (pooled, intercept-shifted, interaction, and separate group models; Section 2) are fitted, and the five stability metrics are averaged across predictors (SSR, $D_j^\star$, $\widetilde{D}_j$) or across groups (the pooled-versus-subgroup prediction gap and CSS). The replicate-level means are then summarized by their mean, standard deviation, and a 95% confidence interval.

3.2 Results

Scenario 1 provides the finite-sample baseline. With identical population slopes and predictor distributions, differences between subgroup estimates arise from random sampling and from realized differences in the subgroup information matrices ($X_g^\top X_g$).

Table 2 gives the main 200-replication comparison. The summaries use the 136 settings per scenario with minimum subgroup size at least three. Medians are used because a small number of boundary settings can produce very large coefficient-drift values.

Table 2: Median diagnostic values across retained settings in the 200-replication simulation study.

Scenario	Drift	Context drift	Prediction gap	CSS	Shrinkage
No heterogeneity	3.64	3.11	0.0043	0.057	5.4%
Covariate shift only	3.78	2.47	0.0044	0.502	33.2%
Coefficient heterogeneity only	24.64	21.74	0.0636	0.059	5.5%
Combined heterogeneity	25.41	16.12	0.0645	0.508	33.5%

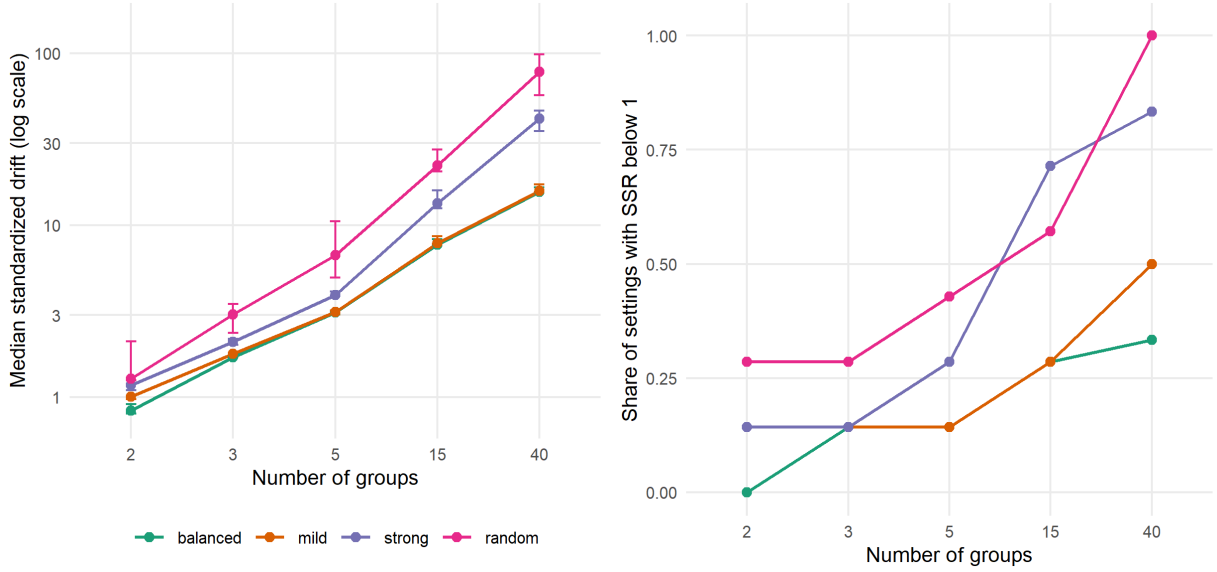


Figure 1: Left: median standardized drift with interquartile range by number of groups and imbalance pattern. Right: share of retained settings with SSR below one. Each point summarizes seven sample-size settings; colours denote imbalance pattern.

3.2.1 Scenario 1 establishes a finite-sample baseline

Figure 1 shows the finite-sample baseline. Drift rises when observations are divided among more groups, and strong or random imbalance increases the rise. On the retained grid, median drift increases from 1.05 at $G = 2$ to 31.14 at $G = 40$. Sign stability stays at one in most settings, but the share of settings with SSR below one rises with the number of groups and reaches one under random imbalance at $G = 40$. Aggregate sample size is therefore not sufficient to determine subgroup stability; the amount and configuration of information within each group also matter.

3.2.2 Covariate shift produces a mechanical signature

Scenario 2 shifts predictor means while retaining common slopes. Its median raw drift is close to the Scenario 1 value (1.04 times as large), whereas its covariate-shift summary is 8.77 times larger. Context normalization reduces median drift by 33.2%, compared with 5.4% in Scenario 1. The pooled-versus-subgroup prediction gap is essentially unchanged from the baseline. In this design, the diagnostics identify substantial distributional separation without evidence that the population slopes have changed.

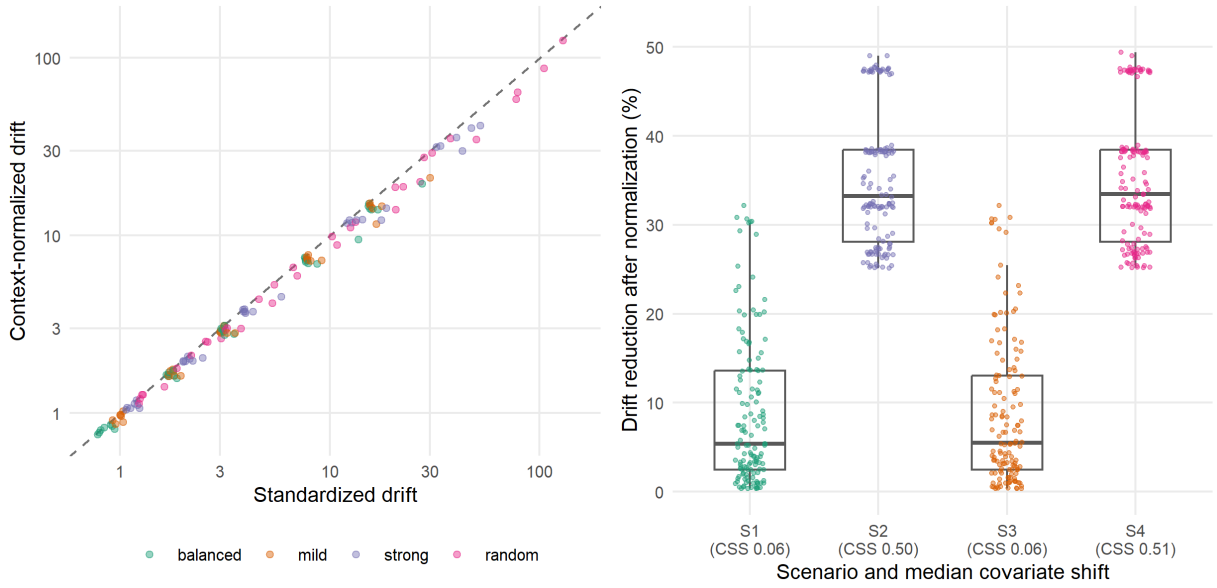


Figure 2: Left: Scenario 1 context-normalized drift against standardized drift on logarithmic axes; the dashed line is equality. Right: distribution of drift reduction after normalization in each scenario, labeled by median covariate shift (S1 no heterogeneity, S2 covariate shift, S3 coefficient heterogeneity, S4 combined). Boxes show medians and interquartile ranges; points are individual settings.

3.2.3 Coefficient heterogeneity creates persistent instability

Scenario 3 changes slopes while keeping predictor distributions comparable. Relative to Scenario 1, median raw drift is 6.76 times larger, context-normalized drift is 6.98 times larger, and the prediction gap is 14.66 times larger. Context normalization removes little of the drift (5.5%). The change in slopes, rather than subgroup mean separation, drives the difference in this design.

Scenario 4 combines shifted means and slopes. It has the largest median raw drift and prediction gap. Relative to Scenario 1, its median raw drift is 6.97 times larger and its prediction gap is 14.87 times larger. It has the lowest mean SSR (0.976), while Scenario 3 has the largest share of retained settings with SSR below one. Its context-normalized drift is lower than in Scenario 3 because the adjustment removes part of the mean-separation component, but it remains 5.18 times the Scenario 1 value.

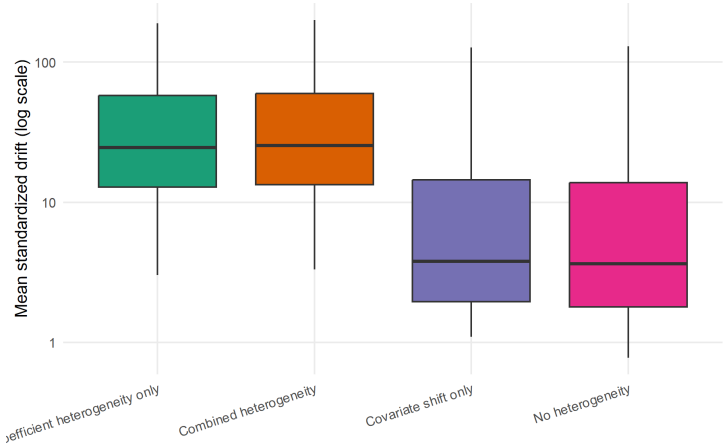


Figure 3: Distribution of mean standardized drift across the 136 retained settings in each scenario. The logarithmic scale emphasizes the separation between the common-slope baseline and settings with coefficient heterogeneity.

3.2.4 Imbalance and group count intensify instability

Balanced and mildly imbalanced designs are generally more stable than strong or random imbalance. Random imbalance gives the highest median drift in Scenarios 1, 2, and 4; strong imbalance does so in Scenario 3. Increasing the number of groups raises drift in every scenario. These patterns show that subgroup comparison becomes less precise both when slopes differ and when the information supporting individual subgroup fits is uneven.

3.2.5 Context normalization distinguishes the two mechanisms

The contrast between Scenarios 2 and 3 is informative. In Scenario 2, large covariate shift accompanies substantial shrinkage after normalization. In Scenario 3, covariate shift remains near the baseline and normalization has little effect, while drift and the prediction gap remain large. Scenario 4 combines these patterns. Context-normalized drift is therefore a descriptive complement to raw drift. It is not a formal decomposition of causal or structural sources of coefficient variation.

3.2.6 Benchmark against the interaction test

Table 3 reports the reduced-grid benchmark of the conventional F-test for group-specific slopes. The test is well calibrated in the two common-slope scenarios: its average rejection rates are 0.051 in Scenario 1 and 0.049 in Scenario 2, close to the nominal 0.05 level. In contrast, the test rejects common slopes in 0.852 of Scenario 3 settings and 0.845 of Scenario 4 settings, averaged across the reduced design grid.

The benchmark confirms that a standard interaction test detects the slope changes built into Scenarios 3 and 4. It does not, however, distinguish Scenario 1 from Scenario 2, nor does it describe the size of coefficient departures, the role of covariate separation, or the within-group predictive cost of pooling. The diagnostics in this paper are intended to supplement, rather than replace, that hypothesis-testing comparison.

Table 3: Average rejection rate of the $\alpha = 0.05$ group-slope interaction F-test across the reduced benchmark grid. Each scenario-setting cell uses 200 replications.

Scenario	Mean rejection rate
No heterogeneity	0.051
Covariate shift only	0.049
Coefficient heterogeneity only	0.852
Combined heterogeneity	0.845

3.2.7 Design-boundary settings

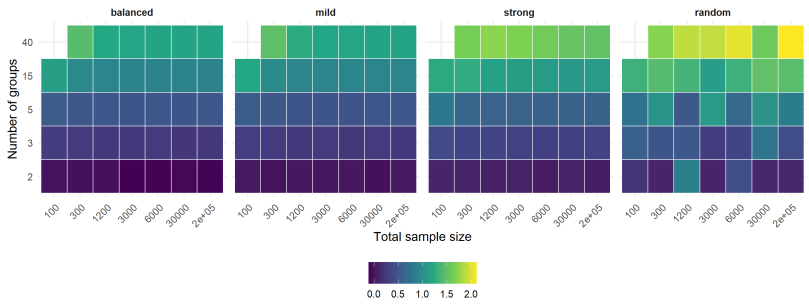


Figure 4: Mean standardized drift as a function of G, total N, and imbalance pattern. Colour indicates $\log_{10}(\text{drift})$. Settings with $\min(n_g) \geq 3$.

Figure 4 displays the baseline pattern across Scenario 1. The high-group, low-sample corner produces extreme estimates when the smallest subgroup has fewer than three observations. These settings are excluded from aggregate comparisons. Medians over the retained grid are therefore more informative than unfiltered means for comparing the scenarios.

4 Empirical Illustration

The empirical illustration uses NHANES data, with age decade as the subgroup variable. Complete cases for AgeDecade, BMI, Gender, Poverty, DirectChol, Pulse, and BPSysAve yield $n = 7116$ observations in eight groups: ages 0 to 9 ($n = 186$), 10 to 19 ($n = 1080$), 20 to 29 ($n = 1126$), 30 to 39 ($n = 1138$), 40 to 49 ($n = 1217$), 50 to 59 ($n = 1116$), 60 to 69 ($n = 782$), and 70 or older ($n = 471$).

The pooled model is

$$\text{BMI} \sim \text{Gender} + \text{Poverty} + \text{DirectChol} + \text{Pulse} + \text{BPSysAve},$$

and the same formula is fitted separately within each age decade. We then compute sign stability, standardized drift, context-normalized drift, and predictor-level summaries.

The aggregate summary shows substantial variation across age groups: mean sign stability is 0.814, mean standardized drift is 16.203, and mean context-normalized drift is 11.339.

Among the predictors, BPSysAve has the lowest sign stability in this specification. Its SSR is 0.571, standardized drift is 32.070, and context-normalized drift is 17.641. Gendermale and DirectChol have SSR equal to one, although DirectChol still has appreciable magnitude drift.

Figure 5 shows the BPSysAve coefficient and its 95% confidence interval in each age group, together with the pooled estimate. The estimated coefficient declines with age and changes sign.

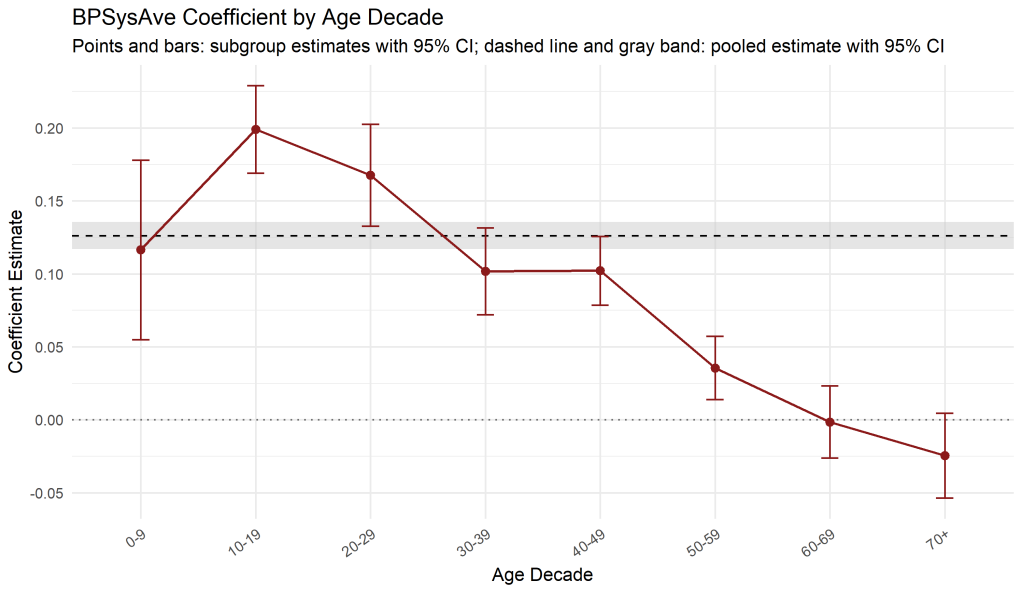


Figure 5: BPSysAve coefficient by age decade with 95% intervals; the dashed line and band show the pooled estimate.

The youngest group is relatively small ($n = 186$), so we repeated the analysis after excluding ages 0 to 9. Several drift values decreased, but the BPSysAve result remained: drift fell from 32.070 to 28.696 and SSR changed from 0.571 to 0.524. The instability is therefore not attributable solely to the smallest group. This sensitivity check does not establish a distinct age-specific mechanism; nonlinear age effects and omitted variables remain possible explanations.

We compared the pooled and age-specific models within each age group. The age-specific model had lower RMSE and higher R^2 in every group. The largest difference occurred for ages 0 to 9, where pooled $R^2 = -3.485$ and subgroup $R^2 = 0.241$. Improvements also remained in the oldest group, where pooled $R^2 = -0.086$ and subgroup $R^2 = 0.130$.

We then conducted 20 repetitions of five-fold cross-validation, assigning folds separately within age groups. Across the 800 held-out group-fold evaluations, the average pooled RMSE exceeded the age-specific RMSE by 0.911. The repeated-CV interval was $[0.813, 1.008]$; this interval describes variation across repeated fold assignments and is not treated as a formal population confidence interval. The pooled model had higher average held-out RMSE in every age group. The largest difference was for ages 0 to 9 (4.461), but positive differences remained in the older groups, including 0.613 for ages 70 and older. Thus, the in-sample pattern is also present in repeated held-out predictions for this specification. Table 4 reports the group-level estimates and their intervals.

Table 4: Repeated five-fold cross-validation by age decade. Positive differences favor the age-specific model; intervals describe variation across fold assignments.

Age decade	Pooled RMSE	Group RMSE	Difference	Interval
0–9	7.777	3.316	4.461	[4.338, 4.584]
10–19	6.266	5.177	1.089	[1.046, 1.132]
20–29	6.233	6.189	0.044	[0.027, 0.060]
30–39	6.540	6.263	0.278	[0.246, 0.309]
40–49	5.971	5.839	0.132	[0.117, 0.147]
50–59	6.056	5.839	0.217	[0.199, 0.234]
60–69	6.517	6.064	0.453	[0.420, 0.486]
70+	6.590	5.977	0.612	[0.549, 0.675]

5 Discussion

5.1 Mathematical interpretation

The simulation separates estimation variation under common slopes from actual differences in subgroup slopes. Write the subgroup data-generating process as

$$Y_g = X_g\beta_g + \varepsilon_g, \qquad \mathbb{E}(\varepsilon_g \mid X_g) = 0.$$

The corresponding pooled population target is

$$\beta^{\dagger}_{\text{pool}} = \left(\sum_{g=1}^{G} X_g^{\top} X_g\right)^{-1} \sum_{g=1}^{G} X_g^{\top} X_g \beta_g.$$

When slopes differ, the pooled coefficient is not generally a simple average of the subgroup coefficients. It is a design-weighted compromise. Groups with more observations, greater predictor variation, or different predictor correlations can have greater influence on $\beta^{\dagger}_{\text{pool}}$.

The observed departure can be written as

$$\hat{\beta}_g - \hat{\beta}_{\text{pool}} = \left(\beta_g - \beta^{\dagger}_{\text{pool}}\right) + \left[(X_g^{\top}X_g)^{-1}X_g^{\top}\varepsilon_g - (X^{\top}X)^{-1}X^{\top}\varepsilon\right].$$

The first term is structural heterogeneity. The second is estimation noise, whose size depends on the subgroup information matrix $X_g^{\top}X_g$. Under common slopes, homoscedastic errors, and a fixed design, the variance of the difference is

$$\operatorname{Var}\left(\hat{\beta}_g - \hat{\beta}_{\text{pool}} \mid X\right) = \sigma^2\left[(X_g^{\top}X_g)^{-1} - (X^{\top}X)^{-1}\right].$$

This expression explains Scenario 1. Dividing the data among groups reduces subgroup information and increases the dispersion of separate OLS estimates, even when every group has the same population coefficient vector.

Covariate shift has a narrower implication than a change in slopes. Under correct linear specification and $\mathbb{E}(\varepsilon_g \mid X_g) = 0$, it does not change the conditional expectation of $\hat{\beta}_g$. It can change $X_g^{\top}X_g$, leverage, and estimator precision. Under model misspecification, it can also change the population best linear approximation. The large covariate-shift and normalization values in Scenario 2 should therefore be interpreted as evidence about design and covariate context, not as proof that covariate shift alone changes a structural slope.

5.2 Signs and prediction

Sign instability is most likely when a coefficient is close to zero relative to its standard error. For a positive coefficient under an approximately normal sampling distribution,

$$\Pr\Big(\hat{\beta}_{j,g} < 0 \mid X_g\Big) \approx \Phi\left(-\frac{\beta_{j,g}}{\mathrm{SE}(\hat{\beta}_{j,g})}\right).$$

Small groups, collinear predictors, and uneven allocations increase $\mathrm{SE}(\hat{\beta}_{j,g})$ and make sign reversals more likely even with common slopes. If population subgroup slopes lie on opposite sides of zero, structural heterogeneity provides an additional source of sign disagreement. The present simulation changes slope magnitudes without forcing a population sign reversal. Its low SSR values at sparse settings should therefore be interpreted mainly as finite-sample behavior.

For prediction, compare the pooled coefficient with the target-group coefficient. Ignoring estimation noise, the excess mean squared prediction error of the pooled model in group g is

$$\mathbb{E}_g\Big[(Y - X^\top \beta^\dagger_{\mathrm{pool}})^2\Big] - \mathbb{E}_g\Big[(Y - X^\top \beta_g)^2\Big] = (\beta^\dagger_{\mathrm{pool}} - \beta_g)^\top \Sigma_g (\beta^\dagger_{\mathrm{pool}} - \beta_g),$$

where $\Sigma_g = \mathbb{E}_g(XX^\top)$. The loss depends both on the difference between the pooled and subgroup slopes and on the target group's covariate geometry. This relation motivates the pooled-versus-subgroup prediction gap used in the simulation. It also shows why coefficient drift alone is not a complete measure of predictive adequacy.

5.3 Empirical illustration

The empirical illustration is consistent with the simulation patterns. In the age-stratified model, the BPSysAve coefficient changes direction in older groups and remains unstable after the smallest group is removed. Age-specific models also improve in-group prediction. These results show that the pooled coefficient is not equally descriptive across the age groups in this specification. They do not establish a causal age interaction or a biomedical mechanism.

5.4 Limitations

The simulation is low-dimensional and linear. Its results need not extend directly to nonlinear models, high-dimensional predictors, regularization, missing data, clustered data, or heteroscedastic errors. Context-normalized drift is a descriptive adjustment for subgroup mean separation, not a causal decomposition. It does not account for all covariance, leverage, misspecification, or measurement-error differences. Further, when slopes differ, the pooled target is design-weighted, so a subgroup's departure from the pooled coefficient is not a symmetric measure of every pairwise group difference.

The empirical illustration uses one observational dataset, one specification, and primarily in-sample comparisons. The age groups are meaningful descriptive strata, but the analysis does not identify causal effect modification. Omitted variables, nonlinear age effects, and residual confounding may contribute to the observed patterns. The simulation uses 200 replications per setting, which materially reduces Monte Carlo error relative to the initial exploratory runs, but it does not replace out-of-sample validation, alternative grouping schemes, or additional empirical applications before the diagnostics are used as a general decision rule.

6 Conclusion

This paper evaluates whether a pooled linear regression remains a useful summary after the data are divided into observed subpopulations. The simulation shows that subgroup coefficient differences have more than one source. Small or uneven groups can generate substantial finite-sample disagreement even with common slopes. Shifts in predictor means can inflate raw coefficient differences and alter the precision of subgroup estimates. Slope heterogeneity produces larger persistent departures and a greater loss from using the pooled model within a subgroup.

The findings do not imply that pooled regression should be rejected whenever subgroup estimates differ. The reduced-grid interaction benchmark detects the simulated slope-heterogeneity scenarios at high rates and remains close to its nominal level when slopes are common. The proposed diagnostics answer a different question: they describe the size of subgroup departures, their relationship to covariate separation, and the within-group predictive cost of pooling. This combination supports a more limited practice: report subgroup stability when pooled coefficients are intended to describe several distinct groups, and interpret coefficient drift alongside covariate separation, conventional interaction tests, and within-group prediction performance. The empirical illustration shows how this practice can reveal a pooled model that is not equally informative across age groups.

The framework is descriptive and finite-sample. It is not a new estimator, a causal procedure, or a replacement for interaction models and stratified regression. Its value lies in making the adequacy of pooling an explicit empirical question.

Data and Code Availability

The simulation code, analysis scripts, figures, and nonrestricted numerical outputs are available to the editor and reviewers on request. They will be deposited in a public repository upon publication. NHANES data are obtained through the source provider and remain subject to their original terms.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Declaration of Interest

The author declares no competing interests.

Declaration of AI Assistance

Generative AI tools were used as writing and programming assistants during this project (Deepseek V4 Pro/Flash, GPT-5.6 Terra, Claude Sonnet 5.5). They assisted with R code development, figure-generation workflows, and language editing. The author directed the research questions and study design, reviewed the analysis and simulation code, verified the reported numerical results, and accepts responsibility for the manuscript's claims and conclusions.

References

Richard Berk, Lawrence Brown, Andreas Buja, Edward George, Arun Kuchibhotla, Weijie Su, and Linda Zhao. Assumption lean regression. *The American Statistician*, 73(S1):119–129, 2019. doi: 10.1080/00031305.2018.1527253.

Dimitris Bertsimas and Ivan Paskov. Stable regression: On the power of optimization over randomization. *Journal of Machine Learning Research*, 21(230):1–25, 2020.

Kenneth D. Garbade. Two methods for examining the stability of regression coefficients. *Journal of the American Statistical Association*, 72(357):54–63, 1977.

Ruiqi Liu, Zuofeng Shang, Yonghui Zhang, and Qiankun Zhou. Identification and estimation in panel models with overspecified number of groups. *Journal of Econometrics*, 215(2):574–590, 2020. doi: 10.1016/j.jeconom.2019.09.010.

Chingkwong Tsoi and Ran Sun. Common reporting errors in subgroup analysis: a comparison of interaction and stratified regression models. *BMC Medical Research Methodology*, 26:87, 2026. doi: 10.1186/s12874-026-02795-3.

Stefan Wager and Susan Athey. Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association*, 113(523):1228–1242, 2018. doi: 10.1080/01621459.2017.1319839.

Mark W. Watson and Robert F. Engle. Testing for regression coefficient stability with a stationary AR(1) alternative. *The Review of Economics and Statistics*, 67(2):341–346, 1985.